

Michael Zhuang

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Education

Bachelor of Science in Computer Science

University of Maryland, College Park | College Park, MD

May 2026

GPA: 3.7/4.0

Skills

Languages: Java, Python, JavaScript, SQL, HTML, CSS

Libraries & Frameworks: React, Spring Boot, FastAPI, scikit-learn

Databases & Messaging: MySQL, PostgreSQL, Redis, Kafka, RabbitMQ, Elasticsearch

Cloud & DevOps: AWS, Docker, Git, Maven, Postman

AI & Machine Learning: Hugging Face, RAG, Chroma, LangChain, PyTorch

Experience

Undergraduate Research Assistant

University of Maryland, College Park

May 2024 - Aug 2026

College Park, MD

Mentored by Dr. Xuanzi Qin, Professor of Health Policy and Management

- Examine large datasets using statistical software and programming languages like Python and SAS, identifying key trends and insights to inform decision-making.
- Participate in team meetings and contribute to project planning and execution, ensuring alignment with technical requirements and team objectives.

Technology Assistant

University of Maryland, College Park

Sep 2023 - May 2026

College Park, MD

- Assist in troubleshooting technical issues and deliver support to end-users, demonstrating strong problem-solving skills and attention to detail.
- Perform research on emerging technologies and best practices in software development, sharing insights and recommendations for improving existing systems and methodologies.

Projects

Food To Eat — Full-Stack AI Food Recommendation Platform

Tech Stack: React, Spring Boot, FastAPI, MyBatis, PostgreSQL, Redis, LangChain

- Developed a full-stack food recommendation platform using React and Spring Boot, combining an interactive frontend with RESTful backend services for food discovery and preference-based recommendations.
- Designed a category-based recommendation algorithm with database-driven filtering to narrow food options through interactive yes/no questions and dynamically refine recommendations.
- Built a conversational AI chatbot using FastAPI, LangChain, and Google Gemini, incorporating Redis-backed conversation management to support natural-language food recommendations.

Food Delivery Platform — Backend Development

Tech Stack: Java, Spring Boot, MyBatis, MySQL, Redis, Maven, AWS S3

- Developed a modular Java backend using Spring Boot and Maven, organizing shared utilities, data models, and application services into separate modules to support maintainable backend development.
- Configured MyBatis and MySQL integration for relational data persistence, establishing the foundation for database-driven food delivery operations and RESTful backend services.
- Integrated dependencies for Redis caching and AWS S3 object storage into the backend application, supporting the implementation of caching and file management capabilities.